

VPA ICT — Grades 2–3 Term 3: Safety & Algorithms

Term 3: Safety & Algorithms — Weeks 25 – 36

Theme: Online safety, unplugged coding, digital citizenship | Duration: 40 minutes per lesson

Term Overview

The final term combines critical digital citizenship concepts with foundational computational thinking. Students learn to stay safe online, protect personal information, and understand basic algorithmic thinking through unplugged activities and introductory block-based coding with Scratch Jr.

Week-by-Week Scheme of Work

Week 25 — Online Safety Rules

Keywords: safe 安全 (ānquán), rule 规则 (guīzé), trusted adult 值得信赖的成年人 (zhídé xìnài de chéngnián rén), password 密码 (mìmǎ), personal information 个人信息 (gèrén xìnxī)

Activity 1 (15 min): "Stop, Think, Click" Song — Students learn a call-and-response song: "Before I click, I stop and think! Is this safe? What would my teacher say?"

Activity 2 (15 min): Safety rules sorting — Picture cards show scenarios (clicking on pop-up ads, asking a parent before downloading, sharing address with a game). Students sort into SAFE and NOT SAFE.

Activity 3 (10 min): Class safety pledge — Students trace and sign a "Digital Safety Pledge" poster.

Formative: Can identify 4/6 safe/unsafe scenarios correctly.

Week 26 — Personal Information

Keywords: name 姓名 (xìngmíng), address 地址 (dìzhǐ), phone number 电话号码 (diànhuà hàomǎ), school name 学校名称 (xuéxiào míngchēng), secret 秘密 (mìmi)

Activity 1 (15 min): "Keep it Secret" game — Students receive cards with different types of information. They decide: "Keep it secret?" or "OK to share?"

Activity 2 (15 min): "My Online Shield" — Students draw a shield and write 3 things they will NEVER share online.

Activity 3 (10 min): Role-play — "A game asks for your real name. What do you do?" Students practice saying "I need to ask my parent/teacher first."

Formative: Can list 3 pieces of personal information to keep private.

Week 27 — Cyberbullying Awareness

Keywords: cyberbullying 网络霸凌 (wǎngluò bàlíng), kind 友善 (yóushàn), unkind 不友善 (bù yóushàn), report 报告 (bàogào), bystander 旁观者 (pángguānzhě)

Activity 1 (15 min): "Kind vs. Unkind" word sort — Students sort message examples: "You are my friend!" vs. "You are stupid!" into kind/unkind.

Activity 2 (15 min): "What would you do?" scenarios — Students discuss what to do if they see unkind messages: (1) Don't reply, (2) Screenshot/tell, (3) Be kind instead.

Activity 3 (10 min): "Kindness Chain" — Each student writes one kind message on a paper strip; chains are linked and hung in the classroom.

Formative: Can describe what to do if they experience or witness cyberbullying.

Week 28 — Screen Time Balance

Keywords: screen time 屏幕时间 (píngmù shíjiān), balance 平衡 (pínghéng), healthy habits 健康习惯 (jiànkāng xíguàn), exercise 运动 (yùndòng), sleep 睡眠 (shuìmián)

Activity 1 (15 min): "My Day" chart — Students draw a pie chart of their day: school, play, screen time, sleep, eating. Discuss: "Is your screen time too big?"

Activity 2 (15 min): "20-20-20 Rule" — Teach the eye health rule: every 20 minutes, look 20 feet away for 20 seconds. Students practice.

Activity 3 (10 min): "Screen Time Promise" — Students write one promise: "I will turn off screens before ____."

Formative: Can explain the 20-20-20 rule and one healthy screen-time habit.

Week 29 — Introduction to Algorithms

Keywords: algorithm 算法 (suànfǎ), steps 步骤 (bùzhòu), sequence 顺序 (shùnxù), instruction 指令 (zhǐlìng), order 次序 (cìxù)

Activity 1 (15 min): "How to Make a Sandwich" unplugged activity — Teacher demonstrates giving unclear instructions. Students laugh. Then give exact steps. Discuss: "Computers need exact steps!"

Activity 2 (15 min): "Brush Your Teeth Algorithm" — Students draw 4-step picture instructions for brushing teeth in correct order.

Activity 3 (10 min): Partner challenge — One student gives verbal instructions; the other follows them exactly to draw a simple shape.

Formative: Can put 4 steps in the correct order to complete a task.

Week 30 — Sequencing Steps

Keywords: first 首先 (shǒuxiān), next 接下来 (jiē xiàláí), then 然后 (ránhòu), finally 最后 (zuìhòu)

Activity 1 (15 min): "Planting a Seed" — Students arrange 4 picture cards in order: dig hole → put seed → cover with soil → water. Then write sequence words.

Activity 2 (15 min): "Morning Routine" — Students draw and sequence their morning: wake up, brush teeth, eat breakfast, go to school.

Activity 3 (10 min): "Sequence Detective" — Teacher shows mixed-up steps; students rearrange.

Formative: Uses sequence words correctly to order steps.

Week 31 — Loops (Repeating Actions)

Keywords: loop 循环 (xúnhuán), repeat 重复 (chóngfù), again 再一次 (zài yíci), pattern 模式 (móshì)

Activity 1 (15 min): "Clap, Clap, Stomp" — Students perform a pattern: clap-clap-stomp, repeat 4 times. Discuss: "Instead of saying clap 8 times, we say 'repeat 4 times!'"

Activity 2 (15 min): "Dance Loop" — In groups of 4, students create a 4-step dance and repeat it 3 times.

Activity 3 (10 min): Loop worksheet — Students circle the repeated part of a pattern and write how many times it repeats.

Formative: Can identify and describe a repeating pattern.

Week 32 — Conditional Statements

Keywords: if 如果 (rúguǒ), then 那么 (nàme), else 否则 (fǒuzé), condition 条件 (tiáojiàn)

Activity 1 (15 min): "If it's sunny, then..." game — Teacher holds up weather cards. Students respond: "If it's sunny, then play outside. If it's raining, then stay inside."

Activity 2 (15 min): "If/Then Simon Says" — "If Simon says jump, then jump! If Simon doesn't say jump, then DON'T jump!"

Activity 3 (10 min): Worksheet — Students complete 3 "If..., then..." sentences with pictures.

Formative: Can complete simple if/then statements correctly.

Week 33 — Introduction to Scratch Jr

Keywords: Scratch Jr Scratch Jr (Scratch 初级版), block 积木块 (jīmù kuài), sprite 角色 (juésè), stage 舞台 (wǔtái), script 脚本 (jiǎoběn)

Activity 1 (15 min): "Explore Scratch Jr" — Students open the app and tap each block category. Teacher names each color/category.

Activity 2 (15 min): "Make the Cat Move" — Students snap together 3 blue motion blocks and press the green flag.

Activity 3 (10 min): "Add a Character" — Students add a new sprite and make both characters move.

Formative: Can snap blocks together and run a program.

Week 34 — Simple Animations

Keywords: animation 动画 (dòng huà), move 移动 (yí dòng), turn 转向 (zhuǎn xiàng), hop 跳跃 (tiào yuè), say 说 (shuō)

Activity 1 (15 min): "Moving Character Project" — Students create a story where a character enters from the left, moves to the center, and says "Hello!"

Activity 2 (15 min): "Add a Sound" — Students record their own voice saying "Hello!" and add it to the project.

Activity 3 (10 min): Partner testing — Students swap tablets and run each other's programs.

Formative: Program contains movement and speech blocks.

Week 35 — Review & Showcase

Activity 1 (20 min): "Project Gallery Walk" — Students display their Scratch Jr projects on tablets around the room. Classmates visit each station and leave a sticker for projects they like.

Activity 2 (15 min): "What I Learned" reflection — Students draw/write 3 things they learned this year.

Activity 3 (5 min): Class photo with digital work displayed.

Week 36 — End of Year Assessment

Portfolio Review (20 min): Teacher reviews each student's digital portfolio containing:

1 digital art piece

1 typed story

1 Scratch Jr project

Online safety poster

Practical Stations (15 min): Students rotate through 3 stations: (1) Type your name and a sentence, (2) Use mouse to complete a puzzle, (3) Show your Scratch Jr project.

Teacher Interview (5 min): "Tell me one way to stay safe online." "Show me how to make the cat move in Scratch Jr."

Standards Mapping

Weeks	Standard	Description
25 - 28	CSTA 1A-IC-17	Work respectfully and responsibly with others online
25 - 28	CSTA 1A-IC-18	Keep login information private
29 - 32	CSTA 1A-AP-09	Model daily processes by creating and following algorithms
33 - 34	CSTA 1A-AP-10	Develop programs with sequences and simple loops
29 - 34	CCSS. MATH. 2. OA. C. 3	Determine whether a group of objects has an odd or even number of members (pattern recognition)

Differentiation for Computational Thinking

Level	Unplugged Coding	Scratch Jr
Bronze	Physically acts out algorithms with teacher guidance	Uses pre-made scripts; modifies numbers
Silver	Creates 3-step algorithms independently	Creates simple scripts with 3-4 blocks
Gold	Creates 5-step algorithms and debugs peers	Creates multi-sprite projects with loops